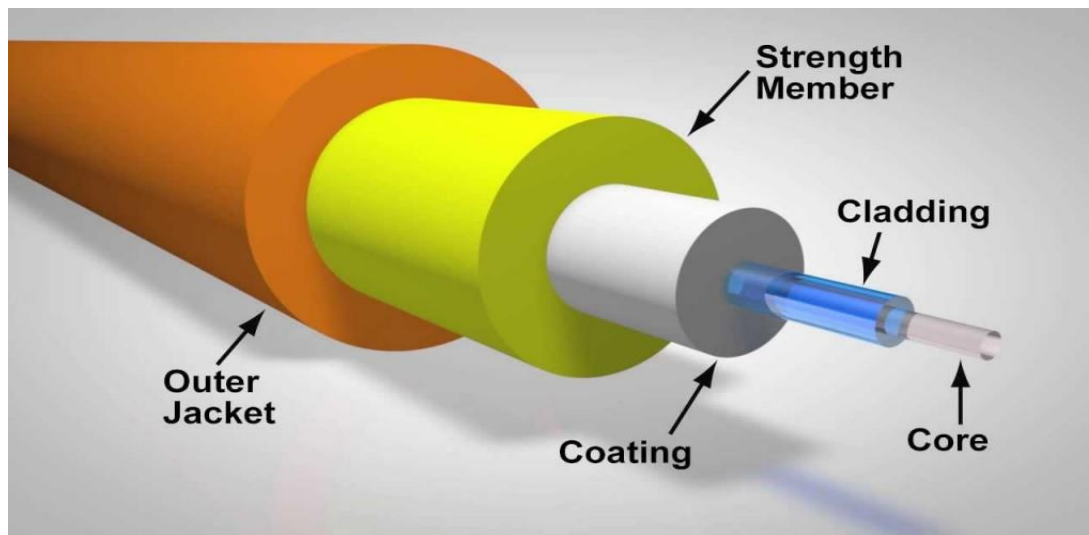


## OPTICAL FIBERS

### I. INTRODUCTION

Optical fibers are essentially light guides used in optical communications as wave guides. They are transparent dielectrics and able to guide visible and infrared light over long distances.

#### Construction:



**Figure-1**

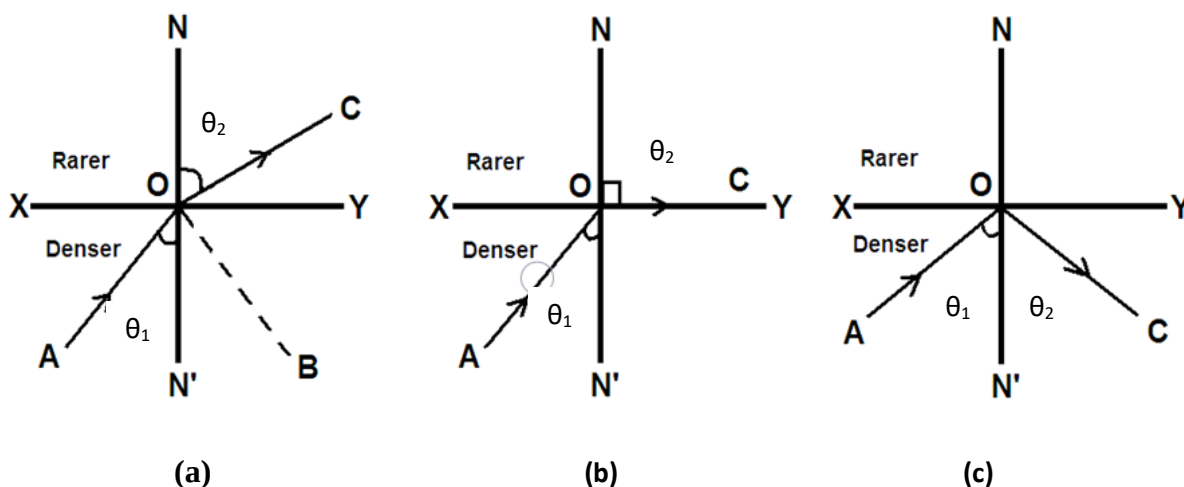
An optical fiber is cylindrical in shape and it has just two parts. The inner part which is a cylindrical material, is made of glass or plastic and is called the *core*. The outer part which is a concentric cylinder surrounding the core is called *cladding*. The cladding is also made of similar material as that of core but of lesser refractive index hence there is a material continuity between core and cladding. The cladding is enclosed in a polyurethane jacket which safe guards the fiber against chemical reaction with surroundings and also against abrasion and crushing. Many such fibers, each one protected by individual jackets are grouped to form a cable. A cable may consist of one to several hundred such fibers.

### II. TOTAL INTERNAL REFLECTION

The physical principle based on which the optical fiber works is the *total internal reflection* which is a well known optical phenomenon in Physics.

Consider a ray AO travelling in a medium of refractive index  $n_1$ . Let XY be the boundary of this medium separating it from another medium of lower refractive index  $n_2$ .

Let the incident ray AO make an angle  $\theta_1$  with the normal in the medium of refractive index  $n_1$ . As this ray is refracted into the medium of refractive index  $n_2$  it bends away from the normal since  $n_1 > n_2$  as shown in figure-2 (a).



**Figure-2**

If  $\theta_2$  is the angle made by the refracted ray with the normal, then  $\theta_2 > \theta_1$ . If  $\theta_1$  is increased then for certain value of  $\theta_1 = \theta_c$  called the *critical angle*,  $\theta_2 = 90^\circ$  the refracted ray just grazes along the boundary of separation along OC while the incident ray is along AO as shown in figure-2(b). For any angle of incidence  $\theta_1$  which is greater than  $\theta_c$  the incident ray like AO always gets reflected back into the medium in which it is incident on the boundary as per the laws of reflection as shown in figure-2(c).

For refraction, we have the *Snell's law*

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

For total internal reflection,

$$\theta_1 = \theta_c, \text{ and } \theta_2 = 90^\circ$$

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

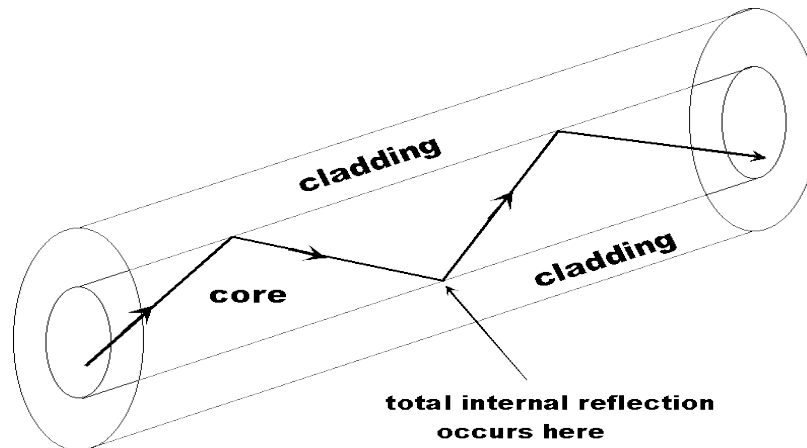
$$\text{Therefore, } \theta_c = \sin^{-1} \left( \frac{n_2}{n_1} \right)$$

### III. PROPAGATION MECHANISM (OR) WORKING PRINCIPLE

A wave guide is a tubular structure through which energy of some sort could be guided in the form of waves. Since light waves can be guided through a fiber it is called *light-guide*. It is also called *fiber wave guide* or *fiber light guide*.

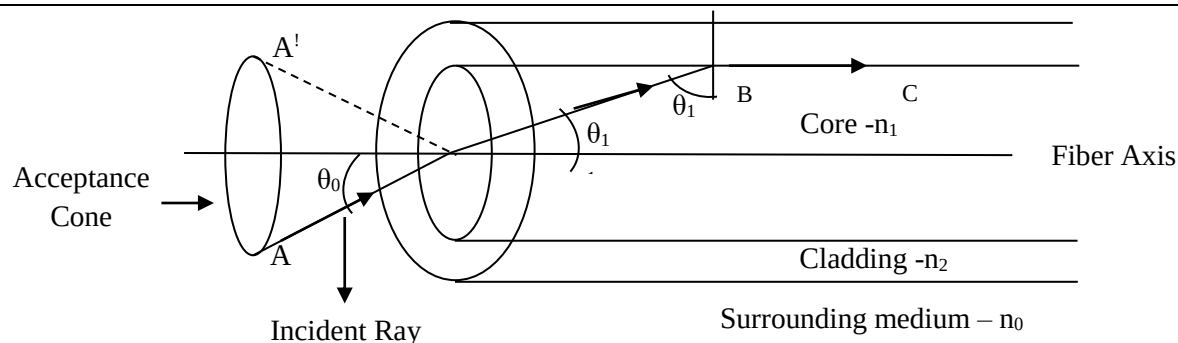
In any optical fiber, the refractive index of cladding is always lesser than that for its core. The light signal which enters into the core can strike the interface of the core and cladding only at large angles of incidence. The signal undergoes reflection after reflection at the point wherever it is incident on the interface. Since each reflection is a total internal reflection, the signal sustains its strength and also confines itself completely within the core during propagation. Thus optical fiber functions as a waveguide.

The propagation of light continues as long as the fiber is not bent too sharply, since for sharp bends the light fails to undergo total internal reflection and the signal strength drops drastically. Hence, care is always taken to avoid very sharp bends in the fiber.



**Figure-3**

#### IV. RAY PROPAGATION IN THE FIBER



**Figure-4**

Let us consider the special case of a ray which suffers critical incidence at the core cladding interface. The ray to begin with travels along AO and entering into the core at an angle  $\theta_0$  to the fiber axis. Then it is refracted along OB at an angle  $\theta_1$  in the core and further proceeds to fall at critical angle of incidence (equal to  $90^\circ - \theta_1$ ) at B on the interface between core and cladding. Since the incidence is critical angle of incidence, the ray is refracted at  $90^\circ$  to the normal drawn to the interface i.e., it grazes along BC.

Any ray that enters into the core at an angle of incidence less than  $\theta_0$ , will have refractive angle less than  $\theta_1$  because of which its angle of incidence (equal to  $90^\circ - \theta_1$ ) at the interface will become greater than the critical angle of incidence and hence undergoes total internal reflection. On the other hand, any ray that enters at an angle of incidence greater than  $\theta_0$  at O, will have to be incident at the interface at an angle less than the critical angle because of which, it gets refracted into the cladding region. Then it travels across the cladding thickness and emerges into the surroundings and thus will be lost.

If OA is rotated around the fiber axis keeping  $\theta_0$  the same then it describes a conical surface. If a beam converges at a wide angle into the core then those rays which are funneled into the fiber within this core will only be totally internally reflected and thus confined within for propagation. Rest of the rays emerge from the sides of the fiber.

The angle  $\theta_0$  is called the *waveguide acceptance angle* and  $\sin \theta_0$  is called the *numerical aperture* (NA) of the fiber. The numerical aperture represents the light – gathering capacity of the optical fiber.

Let  $n_0, n_1$  and  $n_2$  be the refractive indices of surrounding medium, core and cladding of the fiber respectively.

Refraction at the point of entry of the ray AO into the core, by applying the Snell's law then

$$n_0 \sin \theta_0 = n_1 \sin \theta_1 \quad \rightarrow(1)$$

At the point B on the interface, by applying the Snell's law we have,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

The angle of incidence  $\theta_1 = (90 - \theta_1)$ ,  $\theta_2 = 90^\circ$

$$n_1 \sin (90 - \theta_1) = n_2 \sin 90^\circ$$

$$n_1 \cos \theta_1 = n_2$$

$$\cos \theta_1 = \frac{n_2}{n_1} \quad \rightarrow(2)$$

Rewriting equation (1) we have ,

$$\sin \theta_0 = \frac{n_1}{n_0} \sin \theta_1$$

$$\sin \theta_0 = \frac{n_1}{n_0} \sqrt{1 - \cos^2 \theta_1}$$

Substituting for  $\cos \theta_1$  from (2) , we have,

$$\sin \theta_0 = \frac{n_1}{n_0} \sqrt{1 - \left( \frac{n_2}{n_1} \right)^2}$$

$$\sin \theta_0 = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

$$N.A = \sin \theta_0 = \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \quad \rightarrow(3)$$

$$\theta_0 = \sin^{-1} \left( \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \right) \quad \rightarrow(4)$$

If the surrounding medium of the fiber is air then  $n_0 = 1$ , hence:

$$\theta_0 = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

If  $\theta_i$  is the angle of incidence of an incident ray then the ray will be able to propagate

$$\text{If } \theta_i < \theta_0$$

$$\text{or } \sin \theta_i < \sin \theta_0$$

$$\text{or } \sin \theta_i < N.A$$

This is the condition for propagation.

## V. FRACTIONAL INDEX CHANGE : $\Delta$

The fractional index change  $\Delta$  is the ratio of the refractive index difference between the core and cladding to the refractive index of core of an optical fiber.

$$\Delta = \frac{n_1 - n_2}{n_1} \rightarrow (5)$$

## VI. RELATION BETWEEN NUMERICAL APERTURE & $\Delta$

$$(n_1 - n_2) = n_1 \Delta \rightarrow (6)$$

We have ,

$$N.A = \sqrt{n_1^2 - n_2^2}$$

$$N.A = \sqrt{(n_1 + n_2)(n_1 - n_2)} \rightarrow (7)$$

$$\text{Since } n_1 \approx n_2, \quad n_1 + n_2 = 2n_1 \rightarrow (8)$$

Substituting (6) and (8) in (7) we get:  $NA = \sqrt{2\Delta n_1^2}$

$$NA = n_1 \sqrt{2\Delta} \rightarrow (9)$$

An increase in the value of  $\Delta$  increases N.A and thus enhances the light gathering capacity of the fiber. We cannot increase  $\Delta$  to a very large value since it leads to intermodal dispersion which causes signal distortion.

## VII. MODES OF PROPAGATION

On the basis of geometrical optics though it is expected that all such rays which enter into the core at an angle less than the angle of acceptance should travel in the core. The application of Maxwell's equation shows that out of the light that enters into the core within the waveguide acceptance angle only the light waves in terms of certain number of modes will be sustained for propagation in the fiber. The modes of propagation are represented as though they are light rays in number. It can be calculated by a parameter called V – NUMBER.

## VIII. V – NUMBER

The number of modes supported for the propagation of light in the fiber is determined by a parameter called V – NUMBER and it is denoted as V.

$$V = \frac{\pi d}{\lambda} \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \rightarrow (10)$$

If the surrounding media is air then  $n_0 = 1$ , then:

$$V = \frac{\pi d}{\lambda} \sqrt{n_1^2 - n_2^2} \quad (\text{OR}) \quad V = \frac{\pi d}{\lambda} (\text{NA})$$

→(11)

Where, d is the diameter of the core.

$n_1$  and  $n_2$  are the refractive indices of core and cladding respectively

$\lambda$  is the wavelength of light propagating in the fiber.

For  $V \gg 1$ , the number of modes supported by the fiber is given by

$$\text{Number of modes} \cong \frac{V^2}{2} \quad \rightarrow(12)$$

## IX. REFRACTIVE INDEX PROFILE

The curve which represents the variation of refractive index with respect to the radial distance from the axis of the fiber is called the *refractive index profile*.

## X. TYPES OF OPTICAL FIBERS

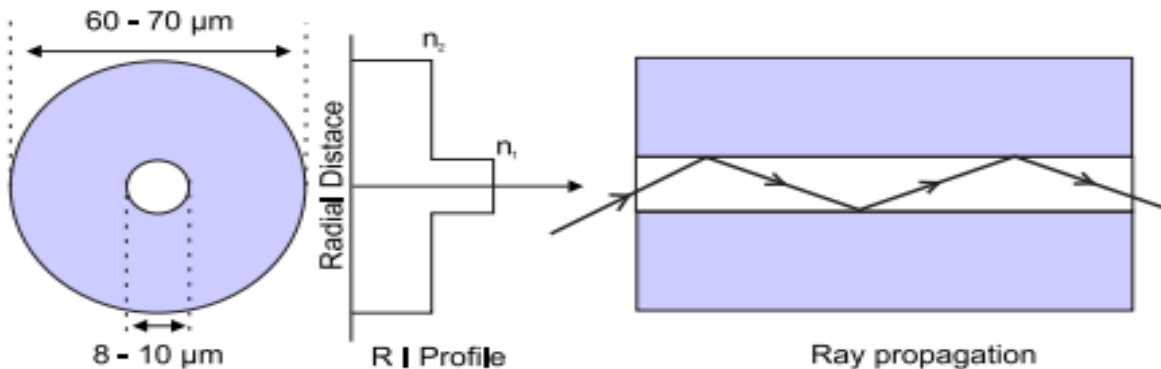
In any optical fiber the whole material of the cladding has a uniform refractive index value. But the refractive index of the core material may either remain constant or subjected to variation in a particular way.

The classification of optical fibers is done on the basis of refractive index profile and the number of modes that the fiber can guide. There are three types of optical fibers they are:

1. Single mode fiber
2. Step index multimode fiber
3. Graded index multimode fiber.

### 1. Single Mode Fiber

A single mode fiber has a core material of uniform refractive index value. Similarly cladding also has a material of uniform refractive index but of lesser value. This results in a sudden increase in the value of refractive index from cladding to core. Thus its refractive index profile takes the shape of a step. The diameter of the core is about  $8 - 10 \mu m$  and external diameter of the cladding is  $60 - 70 \mu m$ .

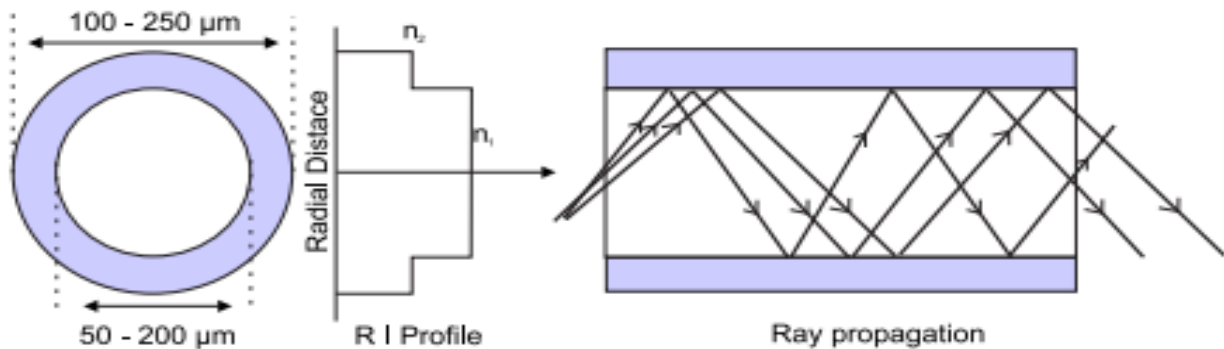


**Figure-5**

Because of its narrow core, it can guide just a single mode. Hence it is called single mode fiber. Single mode fibers are the most extensively used ones and they constitute 80% of all the fibers that are manufactured in the world today. They need lasers as the source of light. Though it is less expensive it is very difficult to splice them. They find particular application in submarine cables system.

## 2. Step index multimode fiber

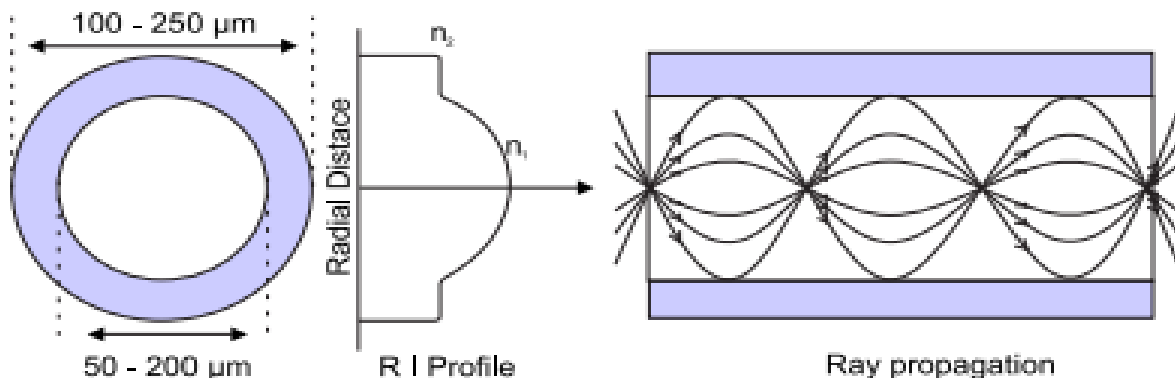
A step index multimode fiber has a core material of uniform refractive index value. Similarly cladding also has a material of uniform refractive index but of lesser value. The diameter of the core is about  $50 - 200 \mu\text{m}$  and external diameter of cladding is about  $100 - 250 \mu\text{m}$ . Here the core has a much larger diameter by the virtue of which it will be able to support propagation of large number of modes as shown in the figure-6.



**Figure-6**

Its refractive index profile is also similar to that of a single mode fiber but with a larger plane region for the core. This fiber can accept either an LED or a laser as a source of light. It is the least expensive of all. Its typical application is in data links which has lower bandwidth requirements.

## 3. Graded Index Multimode Fiber



**Figure-7**

Graded index multimode fiber is also denoted as GRIN. The geometry of GRIN multimode fiber is same as that of step index multimode fiber. Its core material has a special feature that its refractive index value decreases in the radially outward direction from the axis and becomes equal to that of cladding at the interface. But the refractive index of the cladding remains uniform. Either a laser or an LED can be the source for the GRIN multimode fiber. It is the most expensive of all. Its splicing could be done with some difficulty. Its typical application is in the telephone trunk between central offices.

## **XI. ATTENUATION**

Attenuation is the loss of power suffered by the optical signal as it propagates through the fiber. It is also called the *fiber loss*.

The three mechanisms through which attenuation takes place are:

1. Absorption losses
2. Scattering losses
3. Radiation losses

### **1. Absorption losses**

In this case the loss of signal power occurs due to absorption of photons associated with the signal. Photons are absorbed by

- (a) impurities in the silica glass of which the fiber is made of
- (b) intrinsic absorption by the glass material itself.

#### **(a) Absorption by impurities**

The type of impurities that are generally present in fiber glass are the transition metal ions such as iron, chromium, cobalt and copper. During signal propagation when photons interact with these impurities, the electrons (in the impurities) absorb the photons and get excited to higher energy level. Later these electrons give up their absorbed energy either as heat energy or light energy. The re-emission of light energy is of no use since it will usually be in a different wavelength and phase with respect to the signal. Hence it is a loss.

The other impurity which would cause significant absorption loss is the OH (hydroxy) ion which enters into the fiber constitution at the time of fiber fabrication. But the technological breakthrough in the fiber manufacturing has made it possible to restrict the OH content to less than 1 part per billion (ppb) and has reduced the absorption remarkably.

#### **(b) Intrinsic absorption**

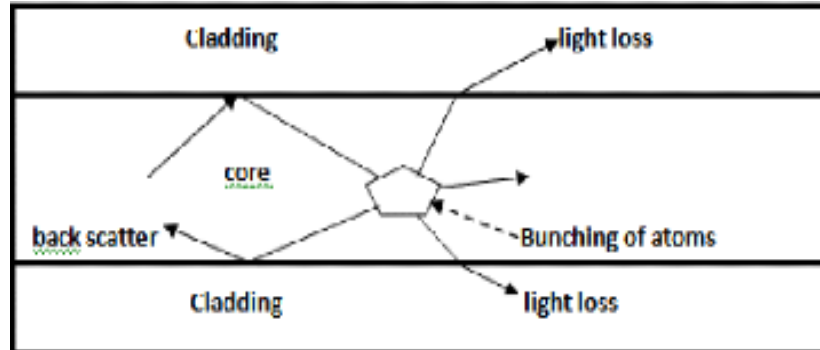
The fiber itself as a material, has a tendency to absorb light energy however small it may be. The absorption that takes place in the fiber material assuming that there are no impurities in it and that the material is free of all inhomogeneities is called *intrinsic absorption*. It sets the lower limit on absorption for a given material.



## 2. Scattering losses

### (a) Rayleigh scattering

Whenever the signal travels in the fiber, the photons may be scattered because of sharp changes in refractive index values inside the glass over distances that are small compared to wavelength of light. The sharp variation in refractive index value inside the fiber glass is induced by the localized structural inhomogeneity. The structural inhomogeneities are set into the glass constitution during its solidification from molten state. This type of scattering is same as Rayleigh scattering.



**Figure-8**

Rayleigh scattering occurs whenever a light wave travels through a medium having scattering objects whose dimensions are smaller than a wavelength. Thus it becomes a loss.

The inhomogeneities in glass cannot be removed by any processing techniques. Thus the loss which depends upon the inhomogeneities becomes a fundamental lower limit on the attenuation.

### (b) Others

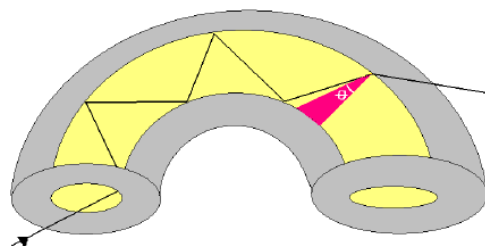
There are defects in the form of trapped gas bubbles unreacted starting materials and some crystallized region in the glass. But the latest manufacturing methods make these losses negligible compared to the Rayleigh scattering.

## 3. Radiation losses

Radiation losses occur due to: (a) macroscopic bends & (b) microscopic bends.

### (a) Macroscopic bends

They are the bends with radii much larger compared to the fiber diameter. These bends occur while wrapping the fiber on a spool or turning it around a corner.



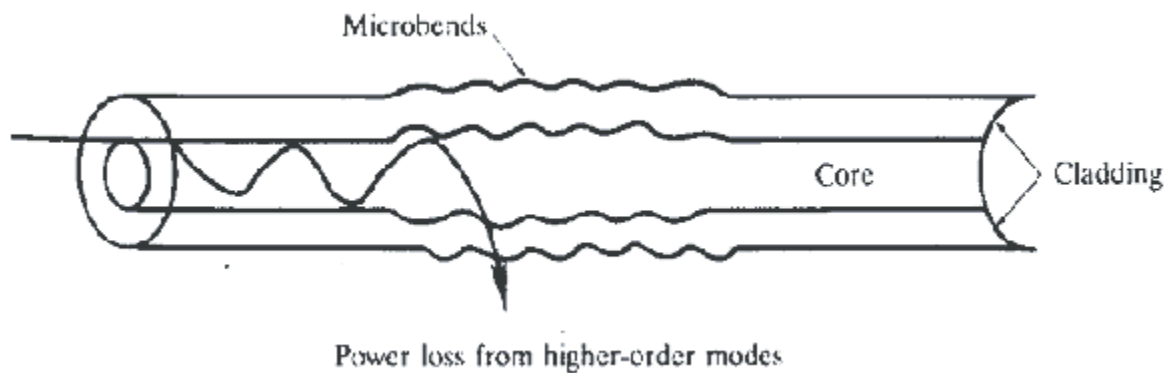
**Macro bend loss.**

**Figure-9**

The loss will be negligible for small bends but increases rapidly until the bending reaches a certain critical radius of curvature. Any bend of radius less than this threshold makes the losses suddenly become extremely large which indicates the absence of total internal reflection.

### (b) Microscopic bends

This type of bends are repetitive small scale fluctuations in the linearity of the fiber axis. They occur due to non uniformities in the manufacturing of the fiber or by non uniform lateral pressures created during the cabling of the fiber. The microbends cause irregular reflections and some of them leak through the fiber. This loss could be minimized by extruding a compressible jacket over the fiber. Then the jacket will be able to withstand the stresses by keeping the fiber relatively straight.



**Figure-10**

## XII. ATTENUATION COEFFICIENT

When light travels in a material medium, there will be always a loss in its intensity with distance travelled. Such a loss of intensity obeys a law called *Lambert's Law*.

As per Lambert's Law, the rate of decrease of intensity of light with distance travelled in a homogeneous medium is proportional to the initial intensity. i.e., if  $P$  is the initial intensity and  $L$  is the distance propagated in the medium, then

$$-\frac{dP}{dL} \propto P$$

(negative sign indicates that it is a decrement)  $-\frac{dP}{dL} = \alpha P \rightarrow (13)$

Where,  $\alpha$  is a constant called the *attenuation coefficient*. It is also called *fiber loss*. It will be different for different media.

Rearrange equation (1) as :  $\frac{dp}{p} = -\alpha dL$

By integrating both sides, we have,  $\int \frac{dP}{P} = -\alpha \int dL \rightarrow (14)$

Considering an optical fiber of length ' $L$ ', if  $P_{in}$  is the initial intensity with which the light is launched into the fiber and  $P_{out}$  is the intensity of the light received as output at the other end of the fiber, then (2) becomes:

$$\int_{P_{in}}^{P_{out}} \frac{dP}{P} = -\alpha \int_0^L dL$$

$$[\ln P]_{P_{in}}^{P_{out}} = -\alpha [L]_0^L$$

$$\ln \left( \frac{P_{out}}{P_{in}} \right) = -\alpha \cdot L$$

$$\alpha = -\frac{1}{L} \cdot \ln \left( \frac{P_{out}}{P_{in}} \right)$$

$$\alpha = -\frac{1}{L} \cdot \log_{10} \left( \frac{P_{out}}{P_{in}} \right) \text{ Bel / Unit length} \rightarrow (15)$$

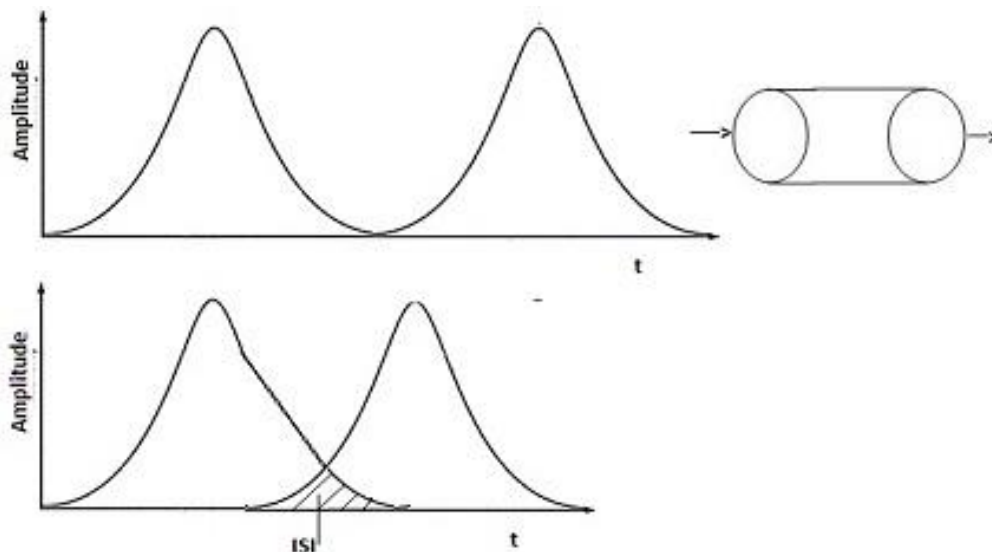
$$1 \text{ Bel} = 10 \text{ deciBel}$$

$$\alpha = -\frac{10}{L} \cdot \log_{10} \left( \frac{P_{out}}{P_{in}} \right) \text{ dB/ km} \rightarrow (16)$$

### XIII. DISPERSION

The spreading of light pulse as it propagates down the fiber is called Dispersion.

This occurs due to the dependence of phase velocity of a wave on its frequency or wavelength. The pulse broadening causes adjacent pulses to get overlapped and limit the maximum number of pulses sent per second, thus reducing the information carrying capacity of the fiber.



**Figure-11**

To avoid overlapping of optical pulses the bit rate (BT) must be less than the reciprocal by broadened duration  $2\tau$ .

$$BT \leq (1/2 \cdot \tau)$$

Two types of dispersion are,

1. Intramodal dispersion.
2. Intermodal dispersion.

### 1. Intramodal Dispersion:

Intramodal Dispersion is also known as ‘chromatic dispersion’ or ‘group velocity dispersion’ (GVD).

When an EM wave travels through a medium of RI, ‘n’ the speed of wave is reduced from speed of light. That is,  $v = c/n$ .

Hence the speed of light in a material depends on its RI which in turn is a frequency dependent parameter. As a result different spectral components of the light pulse travels at slightly different group velocities which causes group velocity dispersion or chromatic dispersion.

Two types of intramodal dispersion are,

- a. *Material Dispersion*
- b. *Waveguide Dispersion*

#### a. *Material Dispersion:*

It is caused by variation of RI as a function of ‘optical wavelength’. Since the group velocity of a mode is a function of RI, the various spectral components of a given mode will travel at different speeds depending on wavelength. The larger the spectral width the higher will be the pulse broadening. Since all wavelength in the spectral width ( $\lambda \pm \Delta\lambda$ ) of the optical source propagate with different speed. It is prominent for LED Sources due to broader spectrum

#### b. *Wave Guide Dispersion:*

This results from the variation in group velocity with wavelength for a particular mode. One mode in the single mode fiber or each mode in a multimode fiber can have its own waveguide dispersion.

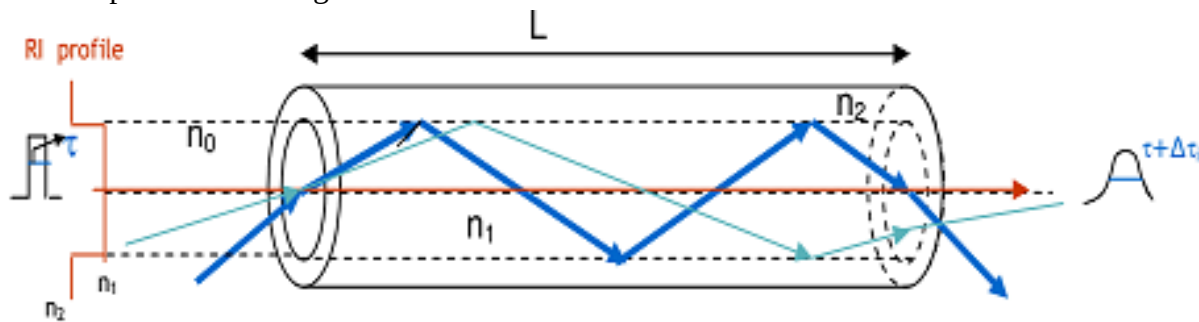
For a mode distribution of light in the fiber varies for different wavelength. Shorter wavelengths are more confined to the core, whereas for larger wavelength, larger portion of power propagates through the cladding. Due to lower RI in the cladding that portion of light travels faster. Also since refractive index depends on wavelength different spectral components in a single mode have different propagation speeds. Thus difference in core-cladding spatial power distribution together with speed variation of various wavelengths causes change in propagation velocity for each spectral component resulting in dispersion.

They are significant in single mode fiber.

### 2. Intermodal Dispersion:

It is also known as modal delay or modal dispersion. Occur due to each mode having different group velocity at a single frequency. Input light is made of a number of modes. Different modes

travel through fibre at different speeds in different directions. Hence, distance travelled by each mode is different. Thus time taken by each mode to cover the same distance is different. This results in pulse broadening.



**Figure-12**

- Increases with distance travelled through the fiber.
- Does not exist in single mode fibers.
- Major source of dispersion is in multimode fibers.

#### **XIV. APPLICATION OF FIBER OPTICS**

##### **Basics of Point to Point Communication System using Optical fibers**

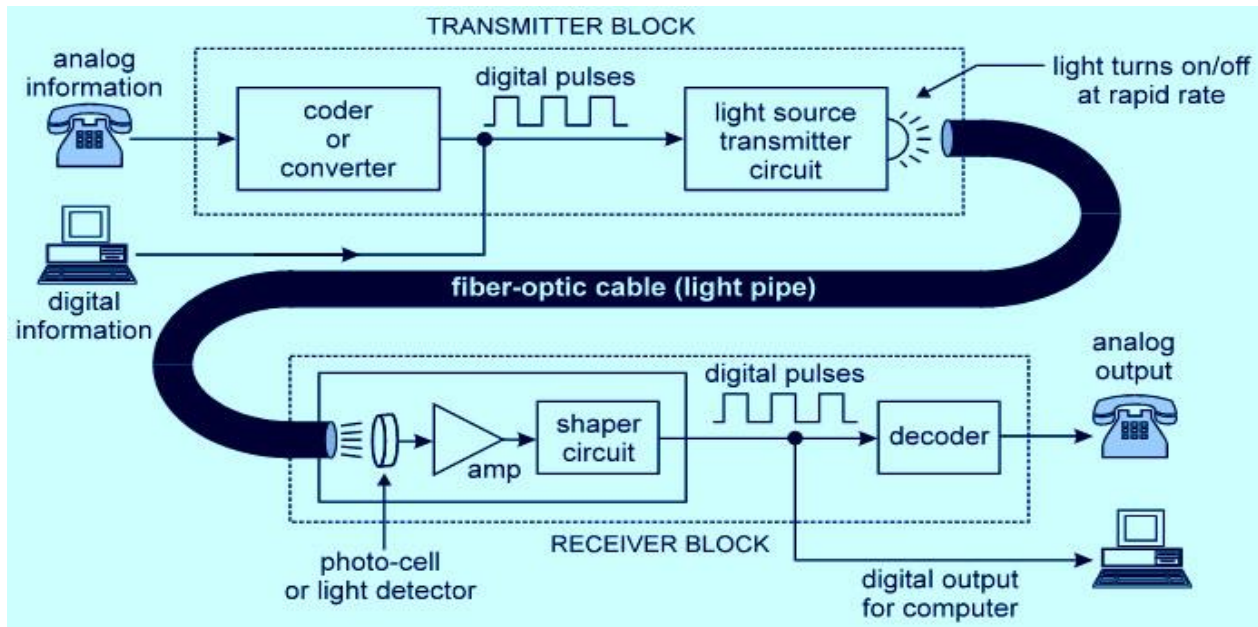
Optical fiber communication is the transmission of information by propagation of optical signal through optical fibers over the required distance which involves deriving optical signal from electrical signal at the transmitting end and conversion of optical signal back to electrical signal at the receiving end.

In a typical point to point communication system, the voice of a telephone user is an analog electrical signal coming out of the transmitter section of the telephone. The analog signal is converted to binary data with the help of an electronic system called *coder*. The binary data comes out as a stream of electrical pulses from the coder. These electrical pulses are converted into pulses of optical power by modulating the light emitted by an optical source in the binary form. This unit is called an *optical transmitter* from which the optical power is fed into the fiber. Out of the incident light which is funneled into the core within the acceptance cone only certain modes will be sustained for propagation within the fiber by means of total internal reflection. As it propagates the signal is subjected to both attenuation and delay distortion. *Delay distortion* is the reduction in the quality of signal because of spreading of pulses with time. The pulse spreading is mainly due to the variation in velocity of various spectral components of the pulse during its propagation in the fiber. Because of such an effect the pulses which are initially separated start overlapping in time.

These effects cause continuous degradation of the signal as the light propagates and may reach a limiting stage beyond which it may not be possible to retrieve the information from the light signal. At this stage a repeater is needed in the transmission path. An *optical repeater* consists of a receiver and a transmitter arranged adjacently. The receiver section converts the optical signal into corresponding electrical signal. Further this electrical signal is amplified and recast in the

original form by means of an electrical regenerator which is also a part of the receiver section. This reshaped electrical signal which is in the form of binary data stream is sent into an optical transmitter section where the electrical signal is again converted back to optical signal and then fed back into the optical fiber.

Finally at the receiving end the optical signal is fed into photo detector where it is transformed into pulses of electric current which is then fed to *decoder* which converts the sequence of binary data stream into an analog signal which will be the same information such as voice which was there at the transmitting end.



**Figure-13**

### Functional Advantages

The functional advantages of optical fibers are –

1. The transmission bandwidth of the fiber optic cables is higher than the metal cables.
2. The amount of data transmission is higher in fiber optic cables.
3. The power loss is very low and hence helpful in long-distance transmissions.
4. Fiber optic cables provide high security and cannot be tapped.
5. Fiber optic cables are the most secure way for data transmission.
6. Fiber optic cables are immune to electromagnetic interference.
7. These are not affected by electrical noise.

### Physical Advantages

The physical advantages of fiber optic cables are –

1. The capacity of these cables is much higher than copper wire cables.
2. Though the capacity is higher, the size of the cable doesn't increase like it does in copper wire cabling system.
3. The space occupied by these cables is much less.
4. The weight of these FOC cables is much lighter than the copper ones.
5. Since these cables are dielectric no spark hazards are present.

6. These cables are more corrosion resistant than copper cables as they are bent easily and are flexible.
7. The raw material for the manufacture of fiber optic cables is glass, which is cheaper than copper.
8. Fiber optic cables last longer than copper cables.

#### Disadvantages

Although fiber optics offer many advantages, they have the following drawbacks –

1. Though fiber optic cables last longer, the installation cost is high.
2. The number of repeaters is to be increased with distance.
3. They are fragile if not enclosed in a plastic sheath. Hence, more protection is needed than copper ones.

#### Applications of Fiber Optics

The optical fibers have many applications. Some of them are as follows –

1. Used in telephone systems and sub-marine cable network.
2. They are used in data link for computer networks, CATV Systems and CCTV surveillance cameras.
3. They have many industrial uses and also used for in heavy duty constructions.